

JOEL SOFFO

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EXPERIENCE	Ph.D CIFRE Candidate <i>AIRBUS</i>	Issy-les-Moulineaux, France Jul. 2023 - Jul. 2026(<i>expected</i>)
	• Supervised by Antoine Bensalah • Research on data-driven methods for solving physics problems. • Research interests: Deep Learning, dimension reduction methods, Physics-Informed Machine Learning, Operator learning, Gradient Descent methods, Optimal Sampling.	
	Machine Learning Scientist <i>AIRBUS</i>	Issy-les-Moulineaux, France Sept. 2022 - Jul. 2023
	Research intern <i>AIRBUS</i>	Issy-les-Moulineaux, France Apr. 2022 - Sept. 2022
EDUCATION	Data Scientist Intern <i>LABELIANS</i>	Nemours, France Apr. 2021 - Sept. 2021
	• Optimization of the tariff base. • Creation of an interface to ease the handing for marketers.	
	Ph.D in Applied Mathematics <i>Ecole Centrale de Nantes</i>	Nantes, France 2023 - 2026 (<i>expected</i>)
PUBLICATIONS	• Advisor: Prof. Anthony Nouy	
	Master Data Science <i>Insitut Polytechnique de Paris (IP Paris)</i>	Palaiseau, France 2021 - 2022
PROJECTS	1. A. Bensalah, A. Nouy, J. Soffo . Nonlinear manifold approximation using compositional polynomial networks. Accepted in <i>SIAM, Journal of Scientific Computing</i> , 2026.	
	2. M. Dus, G. Dusson, V. Ehrlacher, C. Guillot, J. Soffo . Comparison between tensor methods and neural networks for the computation of the Schrodinger equation ground state in electronic structure calculations. <i>ESAIM : Proceedings and Survey</i> , 2024.	
	CEMRACS <i>One month research summer school; Investigation of deep learning methods to solve Schrodinger equation.</i>	2023
	Generative Modelling for Financial Losses <i>AI competition by BNP Paribas and Ecole Polytechnique for the development of stress test methods using deep learning (GANs, VAEs)</i>	2022
	Detection of Ransomware Attack Families via Bitcoin Transactions <i>Features extraction, features analysis, Machine Learning algorithms</i>	2022

AWARDS AND HONORS	• Third Prize , Generative Modeling Data Science Challenge by BNP Paribas / X	2022
	• Best intern of the year , LABELIANS	2021
	• Academic Scholarship , Fondation Mathématique Jacques Hadamard (FMJH)	2020
SKILLS	Languages: French, English.	
	Programming: Python, R, MATLAB.	
	Tools: Git, VS Code, Pycharm.	
	Frameworks: PyTorch, Tensorflow, Keras, Scikit-learn, JAX.	
TALKS AND POSTERS	• (Poster) MORTech 2025 (<i>Zaragoza, Spain</i>) «Nonlinear manifold approximation using CPNs»	
	• (Talk) SMAI 2025 (<i>Bordeaux, France</i>) «Nonlinear manifold approximation using CPNs»	
	• (Talk) CJC-MA 2024 (<i>Lyon, France</i>) «Nonlinear model reduction for solving PDEs»	
	• (Poster) SAMO Conference (<i>Grenoble, France</i>) «Nonlinear dimensionality reduction»	
	• (Talk) Renault 2024 (<i>Paris, France</i>) «Beyond PINNs - Operator Learning»	
	• (Talk) Workshop IRT SystemX 2023 (<i>Palaiseau, France</i>) «Operator Learning methods for solving PDEs»	
TEACHING	• Teaching assistant for first-year undergraduates; Course on «Mathématiques pour les Sciences», <i>Sorbonne Université (Paris-VI)</i>	2023-2024
	• Teaching assistant for second-year undergraduates; Course on series (Numerical series, Function series, Fourier series), <i>ESIEE Paris</i>	2023-2024
MANAGEMENT	Internship supervisor of Takumi HADA (ENSTA 2024) - Now Data scientist at Suntory (<i>Tokyo, Japan</i>) Internship subject : «Machine Learning for simulations».	